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CÔNG NGHỆ KỸ THUẬT TP. HỒ CHÍ MINH
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ARIN330585 – Trí tuệ nhân tạo

Introduction to Artificial Intelligence

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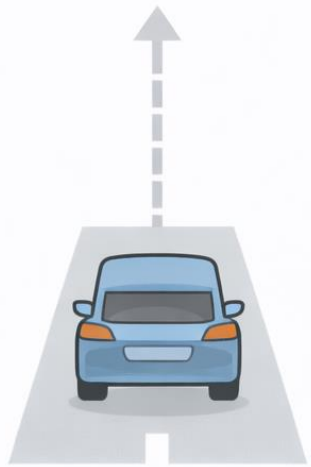
Chapter 7: Search in complex environments

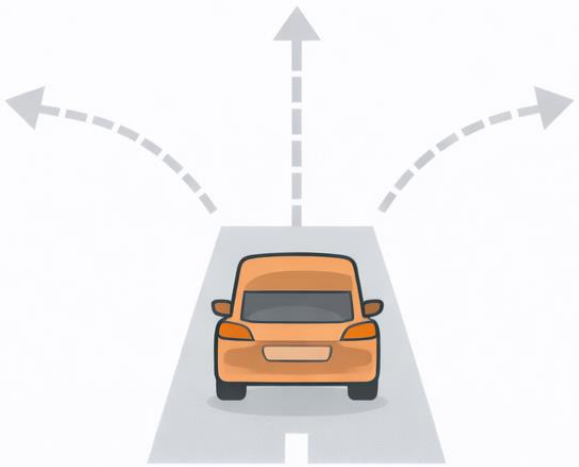
1. *Searching in stochastic environments: AND – OR search*
2. *Searching in unobservable environments*
3. *Searching in partially observable environments*
4. *Online search*

1. Searching in stochastic environments: Environments



DETERMINISTIC

Future outcome is predictable



STOCHASTIC

Future outcome is uncertain

Property 3

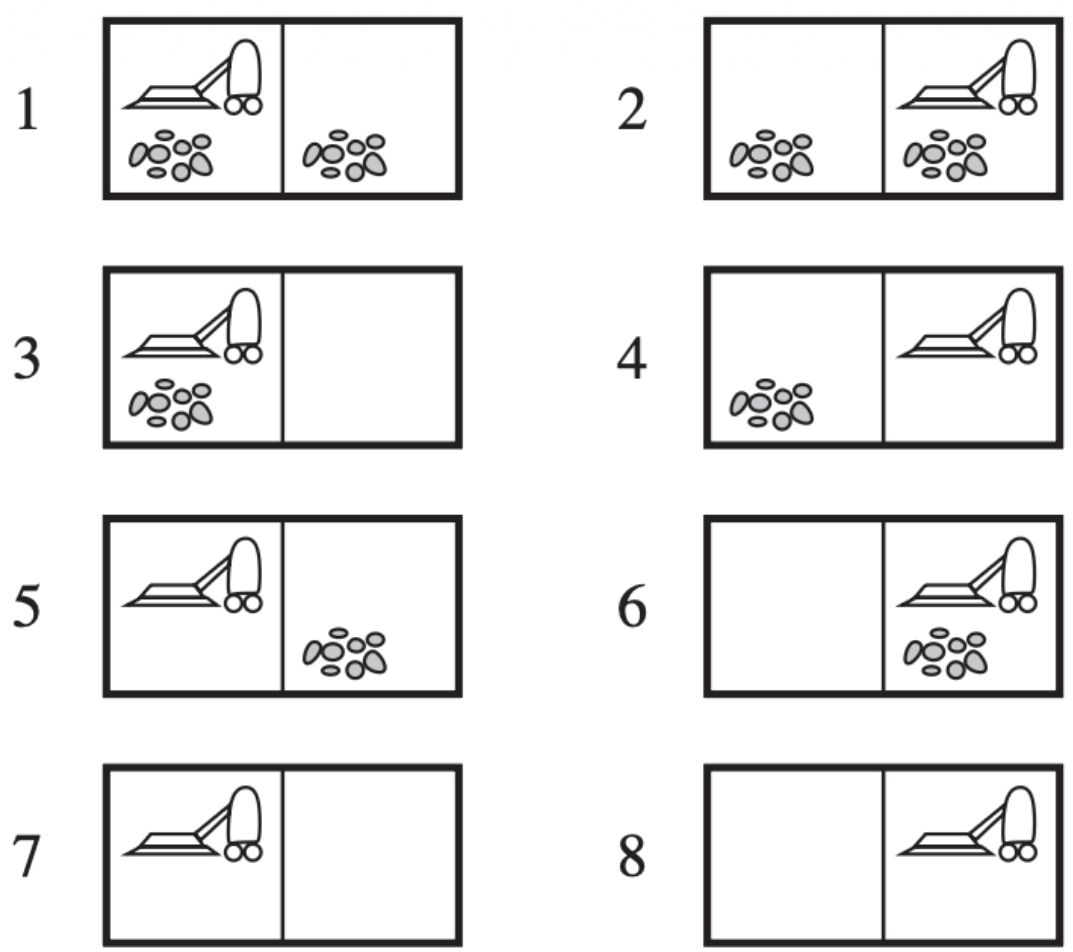
- **Deterministic:** The next state of the environment is completely determined by the current state and the agent's action.
- **Stochastic:** The next state of the environment involves randomness, so the same action can lead to different outcomes.

1. Searching in stochastic environments: Erratic vacuum machine

- Environment: (location, status)
- The state space: 8 states.
- Actions: Left, Right, Suck
- Goal: State 7 and 8.

Consider a ***stochastic environment***:

- When applied to a dirty square the action cleans the square and sometimes cleans up dirt in an adjacent square, too.
- When applied to a clean square the action sometimes deposits dirt on the carpet.



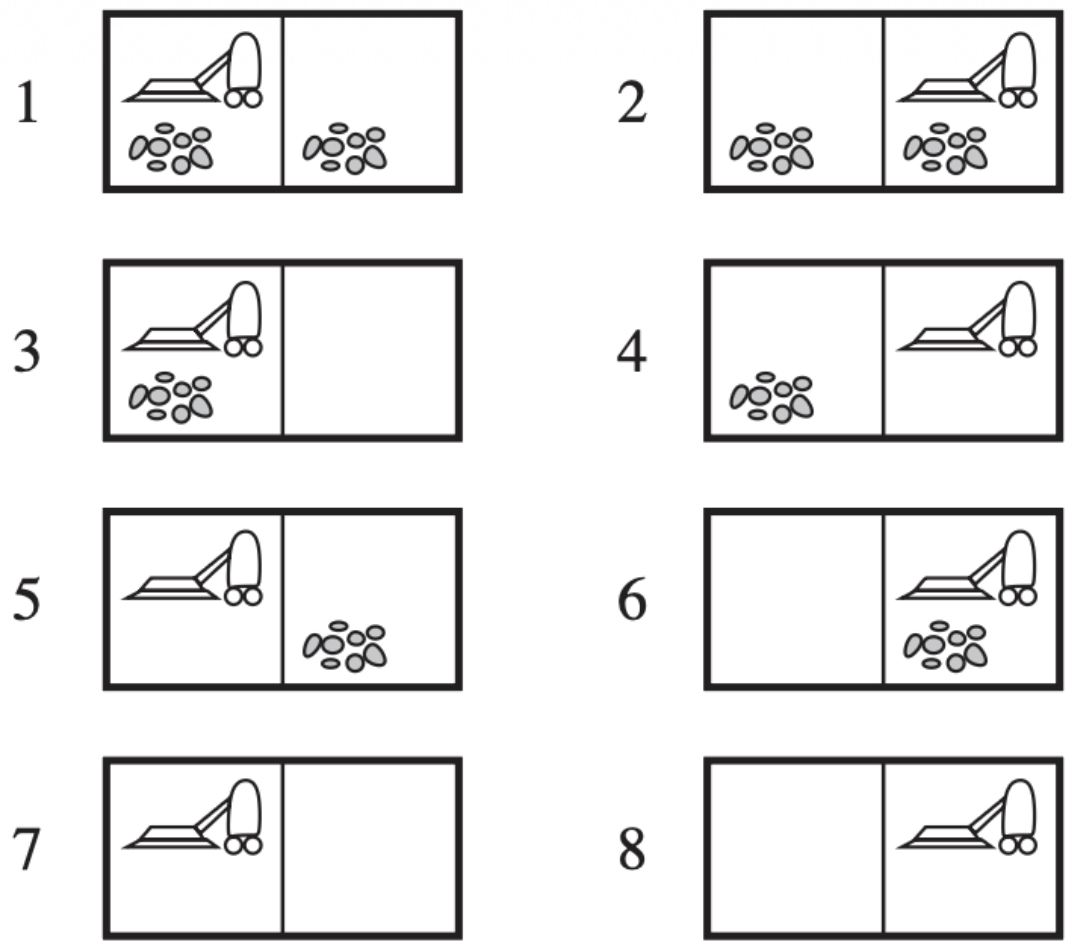
Erratic vacuum machine

- Environment: (location, status)
- The state space: 8 states.
- Actions: Left, Right, Suck
- Goal: State 7 and 8.

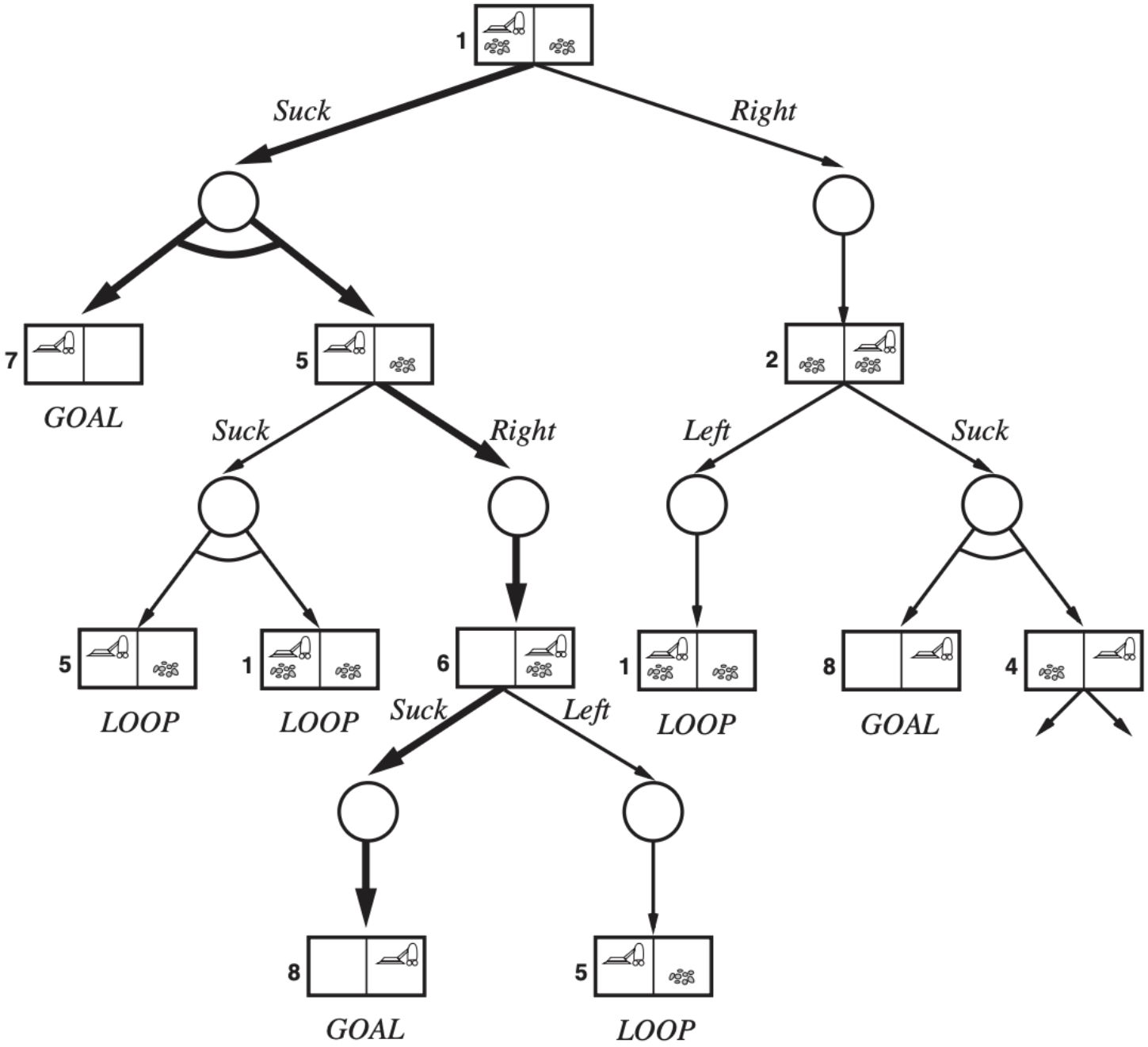
Example: From the state 1, the action Suck can lead to either the state 5 or 7.

We need a contingency plan:

[Suck, **if** State = 5 **then** [Right, Suck] **else** []]



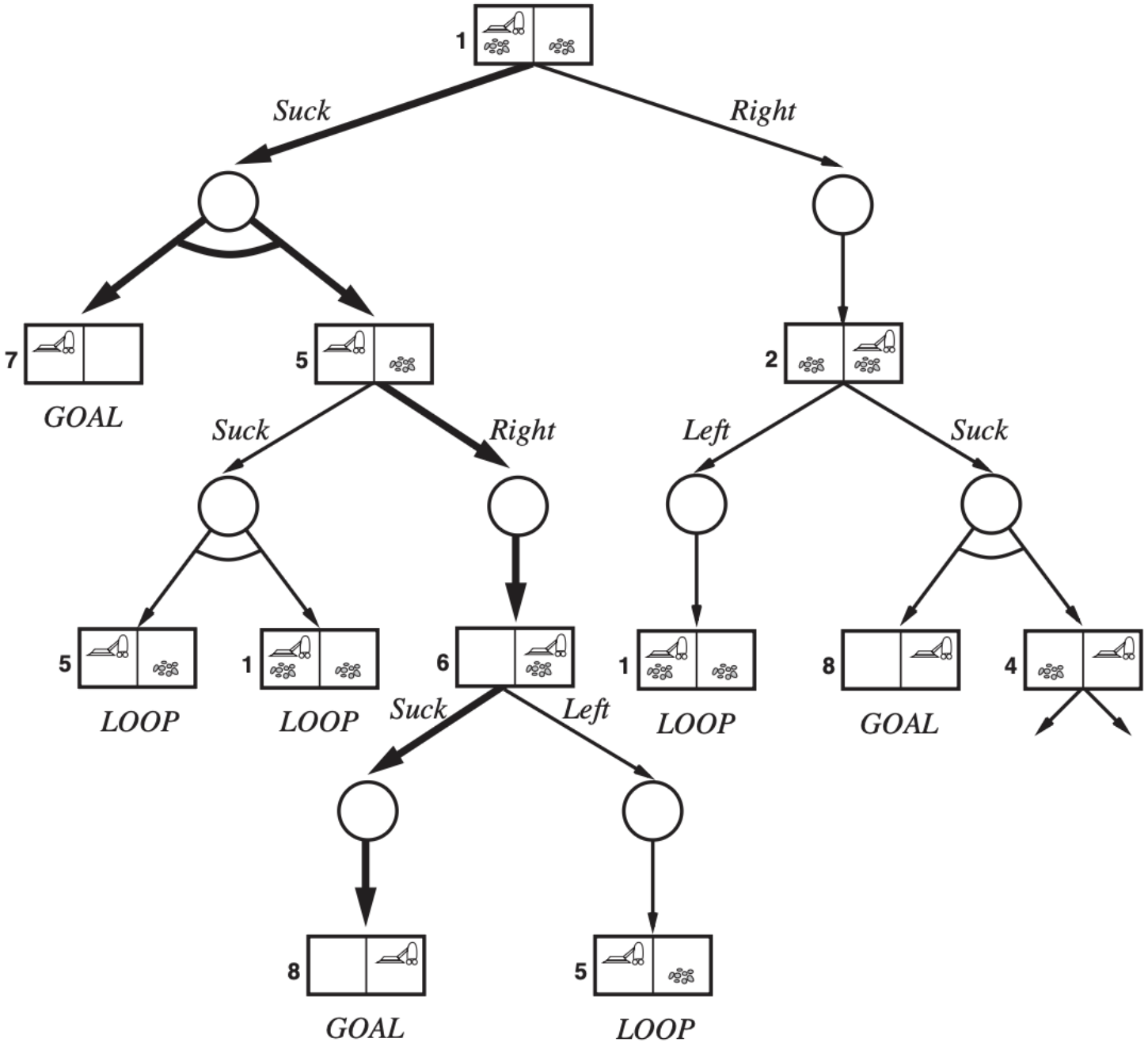
AND – OR search trees



AND – OR search trees:

- **OR nodes** represent the *agent's* own choices.
- **AND nodes** are introduced by the *environment's* choice of outcome.

AND – OR search trees



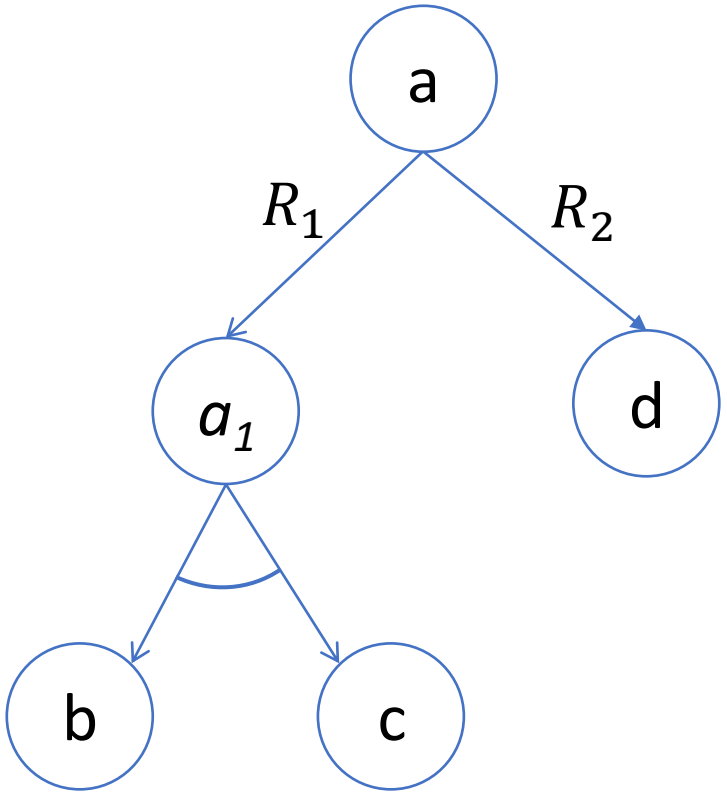
A solution is a **subtree** that:

- (1) has a goal node at every leaf.
- (2) specifies one action at each of its OR nodes.
- (3) includes every outcome branch at each of its AND nodes.

From trees to graphs

How to build an AND-OR graph:

- A node represents for each state
- If an action R transforms one state into another, e.g. R: a → b, then there is an edge from a to b.
- If R transforms one state into multiple substates, e.g. R: a → (b, c), then we introduce a new node a_1. This node represents the set of substates b, c associated with the action R.



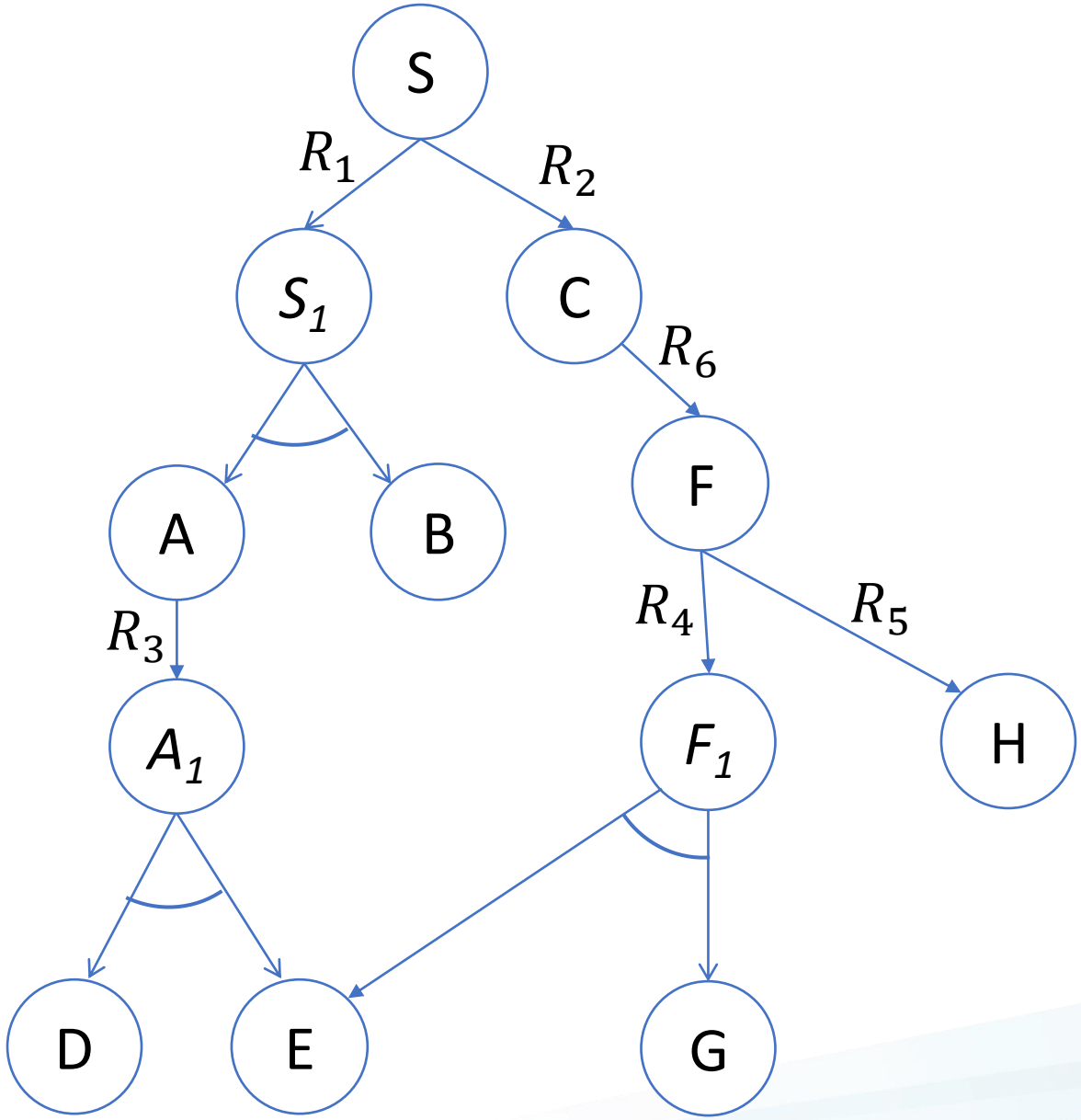
From trees to graphs

Example: Build the AND-OR graph of the following problem:

Initial state: S

Set of actions:

- R1: S → A, B
- R2: S → C
- R3: A → D, E
- R4: F → G, E
- R5: F → H
- R6: C → F



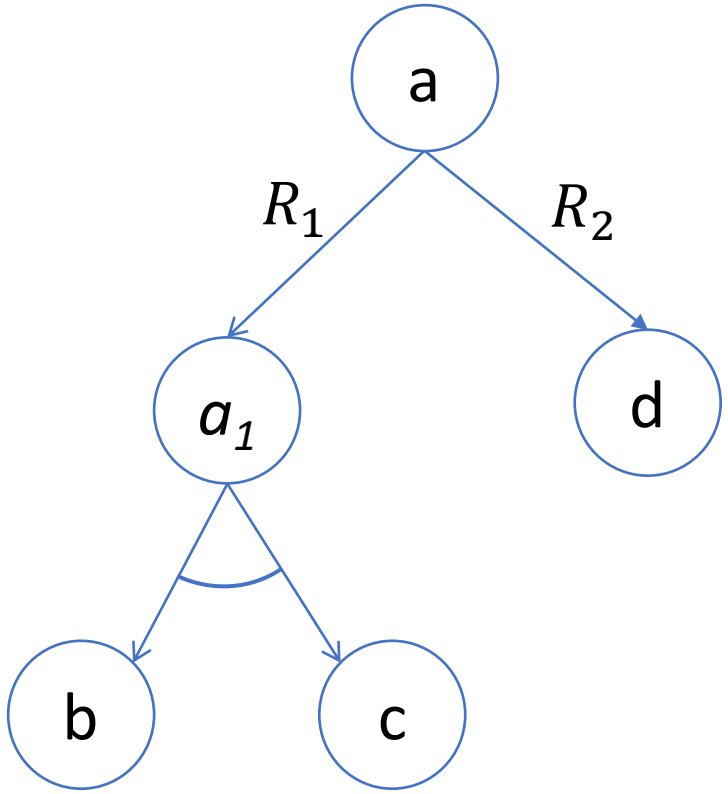
AND-OR graph search

We usually use the DFS algorithm to solve the problem.

```
function AND-OR-GRAPH-SEARCH(problem) returns a conditional plan, or failure
    OR-SEARCH(problem.INITIAL-STATE, problem, [])

---

function OR-SEARCH(state, problem, path) returns a conditional plan, or failure
    if problem.GOAL-TEST(state) then return the empty plan
    if state is on path then return failure
    for each action in problem.ACTIONS(state) do
        plan ← AND-SEARCH(RESULTS(state, action), problem, [state | path])
        if plan ≠ failure then return [action | plan]
    return failure

---

function AND-SEARCH(states, problem, path) returns a conditional plan, or failure
    for each s_i in states do
        plan_i ← OR-SEARCH(s_i, problem, path)
        if plan_i = failure then return failure
    return [if s_1 then plan_1 else if s_2 then plan_2 else ... if s_{n-1} then plan_{n-1} else plan_n]
```

AND-OR graph search

For example, consider the problem bellow where b and c are goals.



AND-OR-GRAPH-SEARCH(problem)
→ OR-SEARCH(a, problem, [])

OR-SEARCH(a, problem, [])
└— try R1
  └— AND-SEARCH({a1}, problem, [a])
    └— OR-SEARCH(a1, problem, [a])
      └— AND-SEARCH({b, c}, problem, [a1, a])
        ├— OR-SEARCH(b, problem, [a1, a]) = empty plan
        └— OR-SEARCH(c, problem, [a1, a]) = empty plan
      = success
    = success
  = success
= success

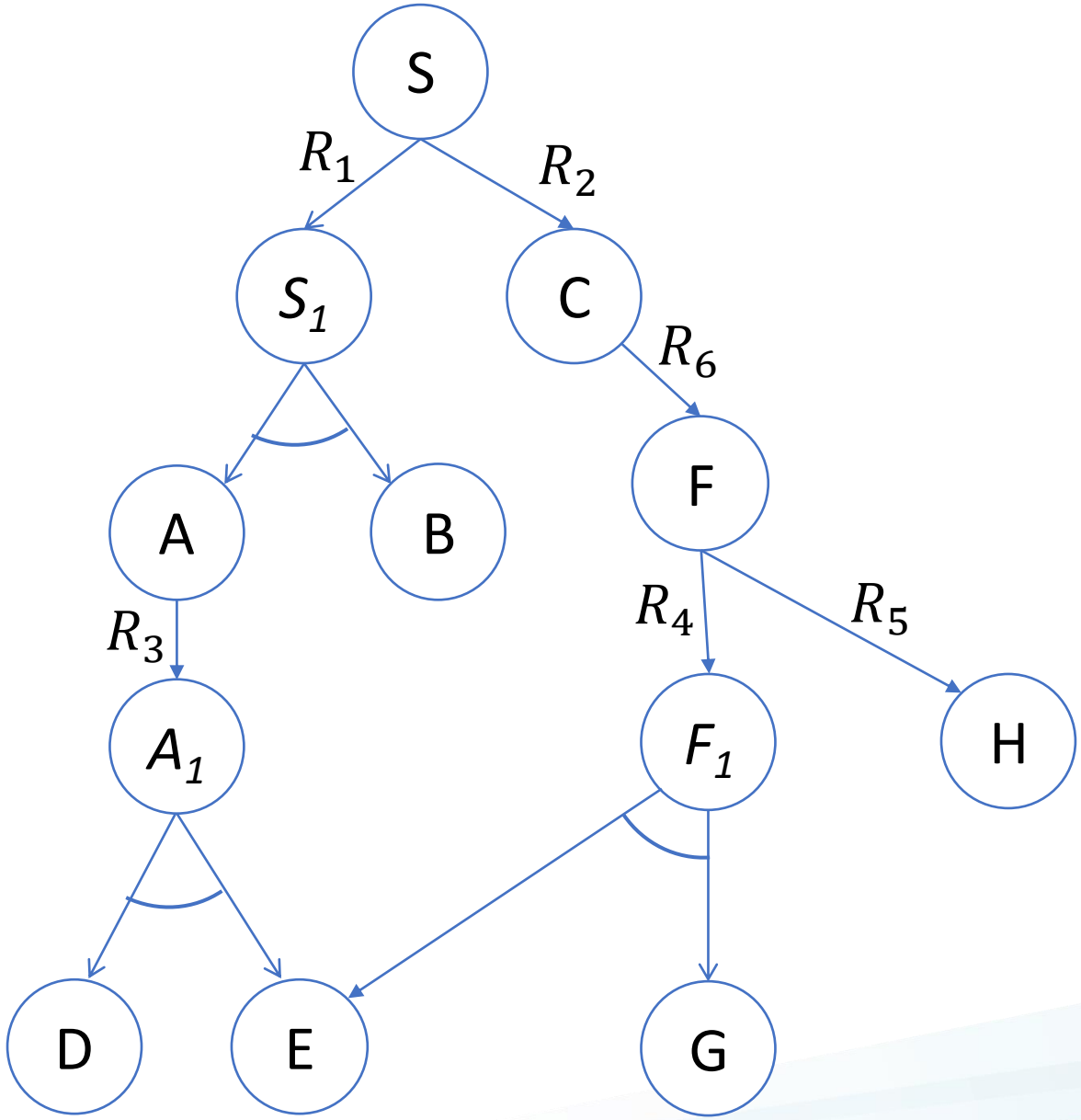
AND-OR graph search

If goal = {B,D,E,G}, is the problem solvable?

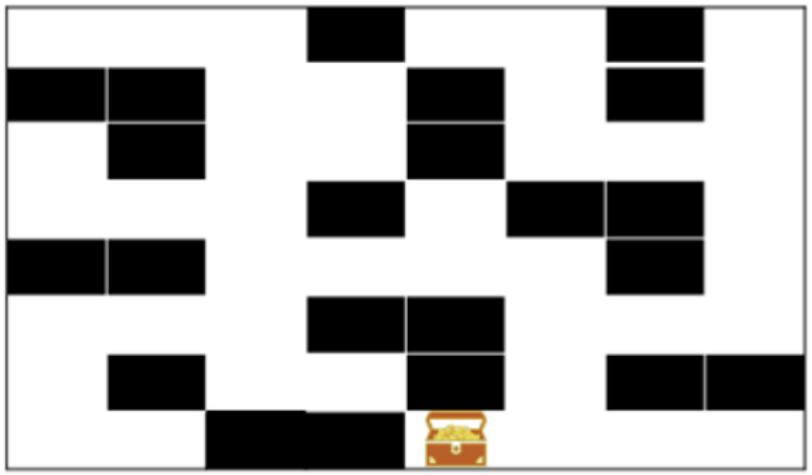
What if goal = {D,E}?

What if goal = {B,D,E}?

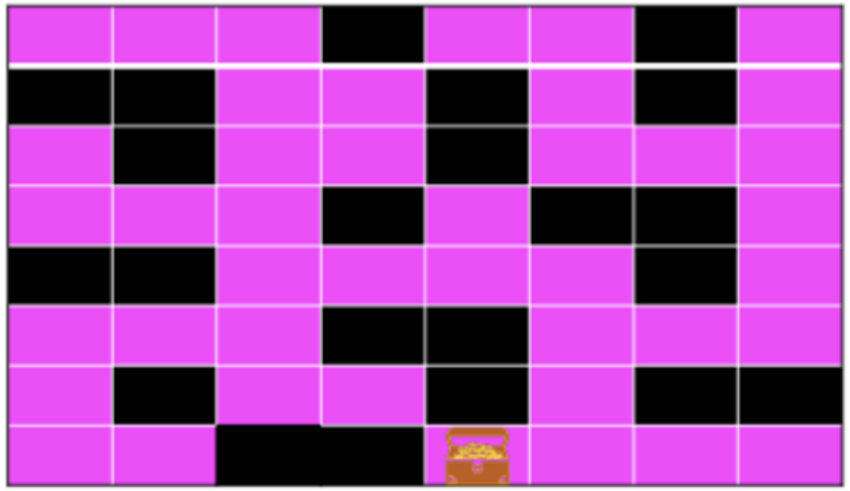
Solution = [
R1,
if state = A then R3,
if state = D then done
else if state = E then done
else if state = B then done
]



2. Searching in unobservable environments: Environments



If the environment is observable, we can solve this problem by using A* algorithm.

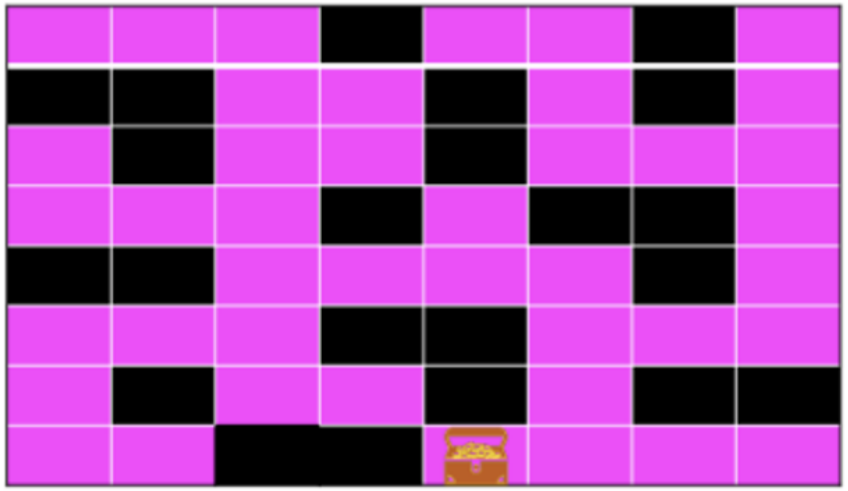


In an unobservable environment:

- We know the layout of the map (known environment)
- We don’t know the initial state
- We don’t know if we move successfully or not

Belief state

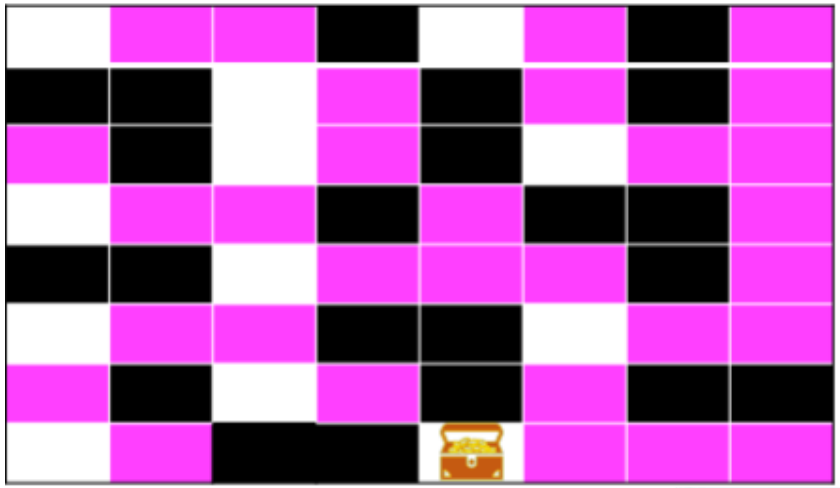
***Belief state*:** represents the agent's current belief about the possible states it might be in, given the sequence of actions and percepts up to that point.



For example:

Initially: The agent doesn't know where it is, then

belief states = all possible squares



After moving right: The agent still doesn't know its location, but it cannot be the white squares.

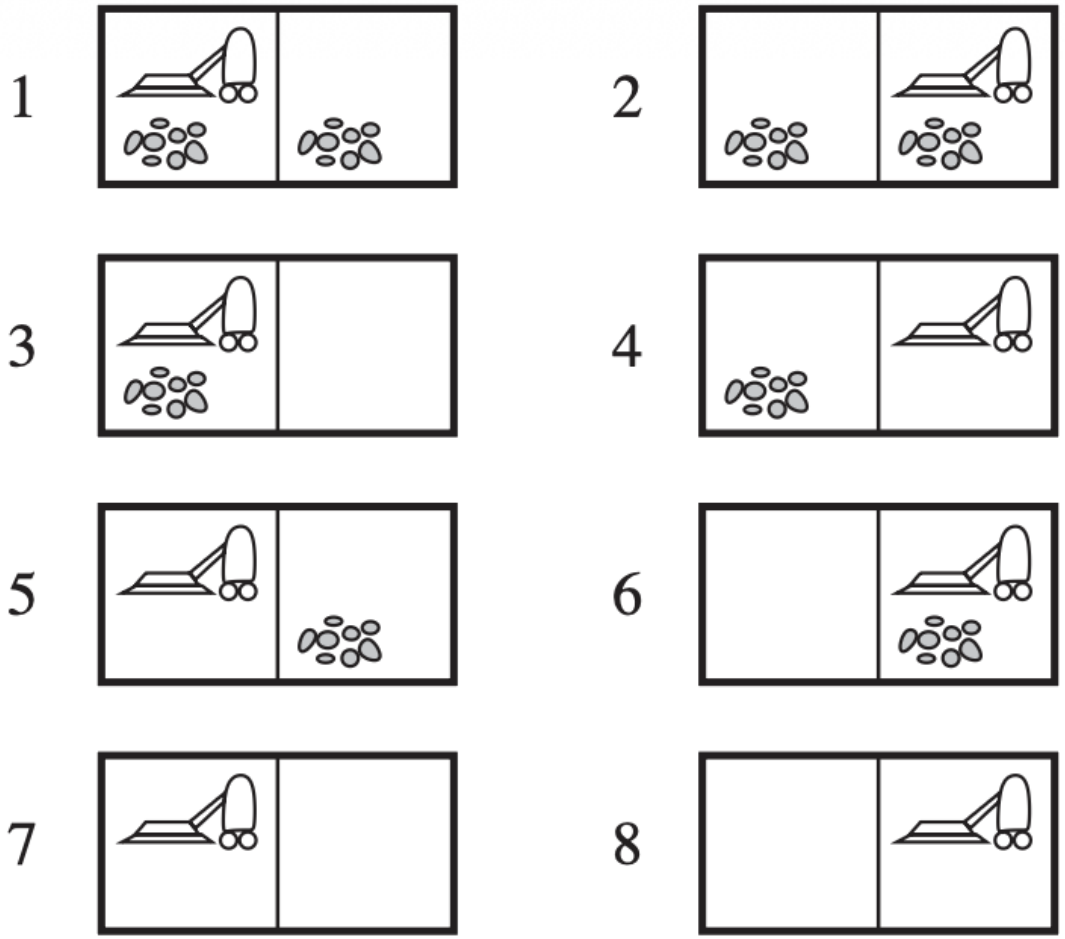
belief states = all possible pink squares

Solutions

Searching without observation = sensorless = conformant.

Example in vacuum world with no sensors:

- Agent knows the layout of the environment.
- Agent doesn't know its location and distribution of dirt
- The solution could be
  - [] → belief states = {1,2,…,8}
  - [Right] → belief states = {2,4,6,8}
  - [Right, Suck] → belief states = {4,8}
  - [Right, Suck, Left] → belief states = {3,7}
  - [Right, Suck, Left, Suck] → belief states = {7}



From physical problem to belief-state problem

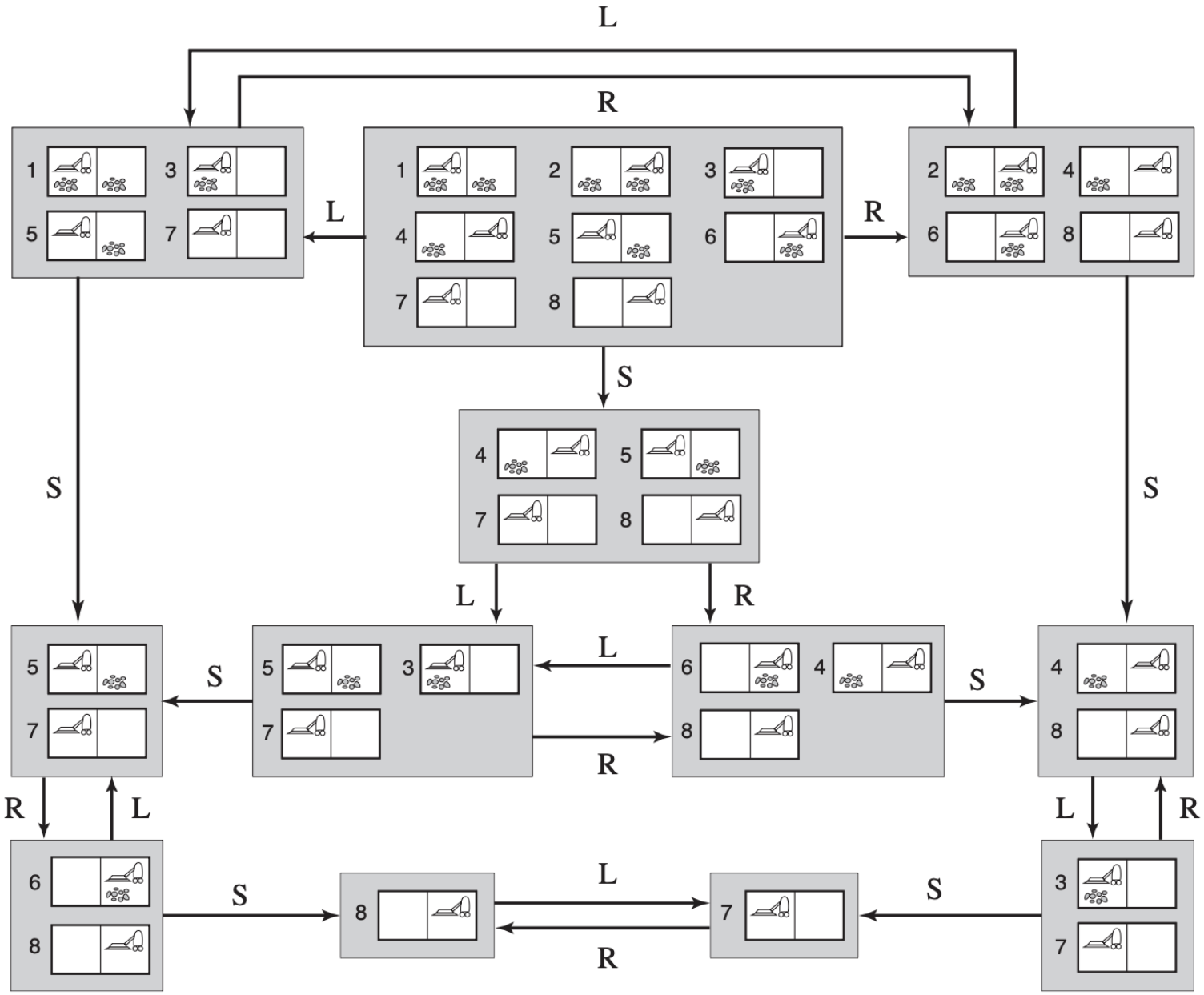
Assume the physical problem P (with N states) is defined by *ACTIONS, RESULT, GOAL-TEST.*

- **Belief states:** every possible set of N state => 2^N states.
- **Initial state:** the set of all states in P.
- **Actions:** typically union of all the actions

  ACTIONS(belief state) = UNION(*ACTIONS*(state))
- **Transition (prediction):**

  b’ = RESULT(b, a) = {s’ : s’ = *RESULT*(s, a) and s ∈ b}
- **Goal test:** a belief state satisfies the goal only if all the physical states in it satisfy *GOAL-TEST.*

Searching in unobservable environments

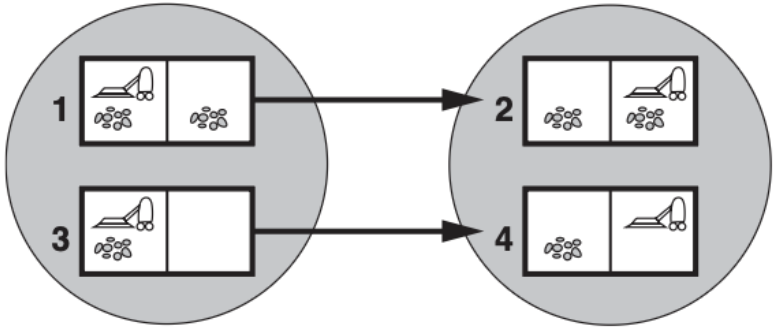


Original (physical) states

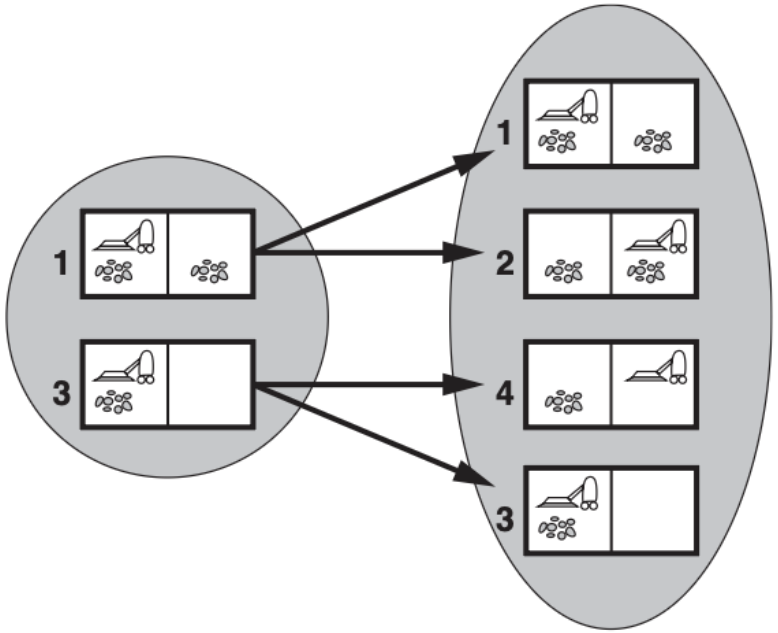
→ reachable belief states

- Reachable belief states = 12, while belief states = 256.
- We can use any search algorithm to solve this problem.

Unobservable and stochastic environments



Predicting the next belief state for the sensorless vacuum world with a deterministic action, *Right*.



Prediction for the same belief state and action in the slippery version of the sensorless vacuum world.

3. Searching in partially observable environments: Environments

Property 1

- **Fully observable**: The agent can see all the relevant aspects of the environment needed to choose the best action.
- **Partially observable**: The agent can see only part of the environment, so it must deal with missing or uncertain information.

Partially observable environments

The formal problem specification includes a PERCEPT(s) function that returns the percept received in a given state.

- Fully observable environments:

  PERCEPT(state) = env

- Unobservable environments:

  PERCEPT(s) = null

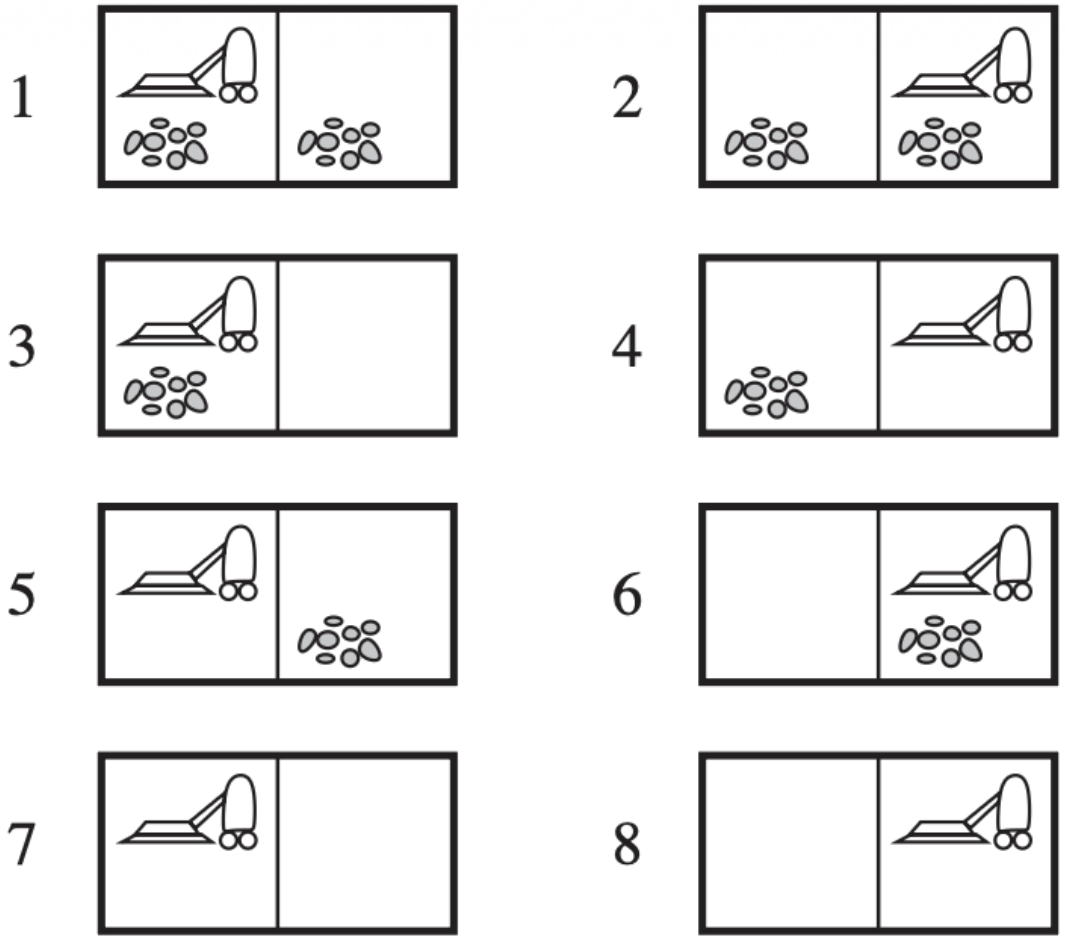
- Partially observable environments:

  PERCEPT(s) = *current percept*

  For example:

  PERCEPT(state 1) = PERCEPT(state 3) = [A, Dirty]

  PERCEPT(state 4) = PERCEPT(state 8) = [B, Clean]



Transition in partially observable environments

The same idea as unobservable environments. However, we now have percept, so the transition step can work more ”correctly.”

- **Prediction:**

$$b' = \mathit{PREDICT}(b,a)$$

- **Observation prediction:**

$$\mathit{POSSIBLE\text{-}PERCEPTS}(b') = \{o: o = \mathit{PERCEPT}(s) \text{ and } s \text{ in } b'\}$$

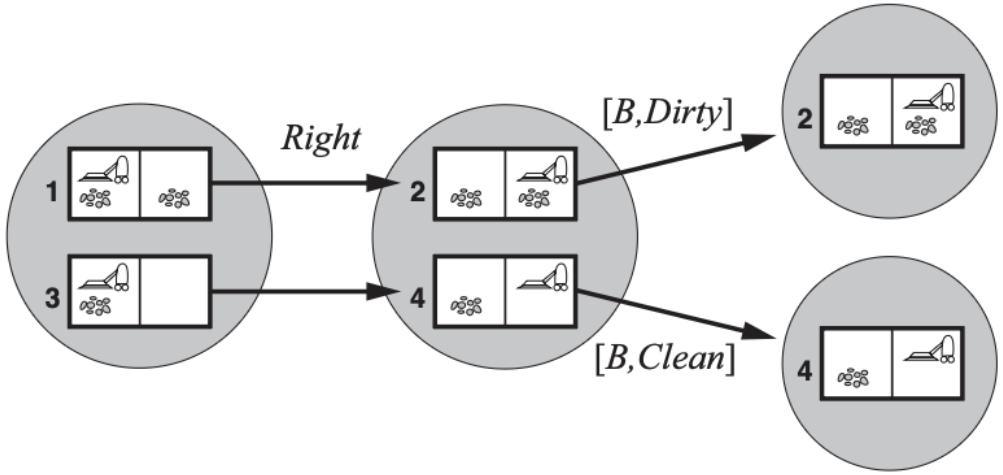
- **Update:**

$$b'' = \mathit{UPDATE}(b',o) = \{s: o = \mathit{PERCEPT}(s) \text{ and } s \text{ in } b'\}$$

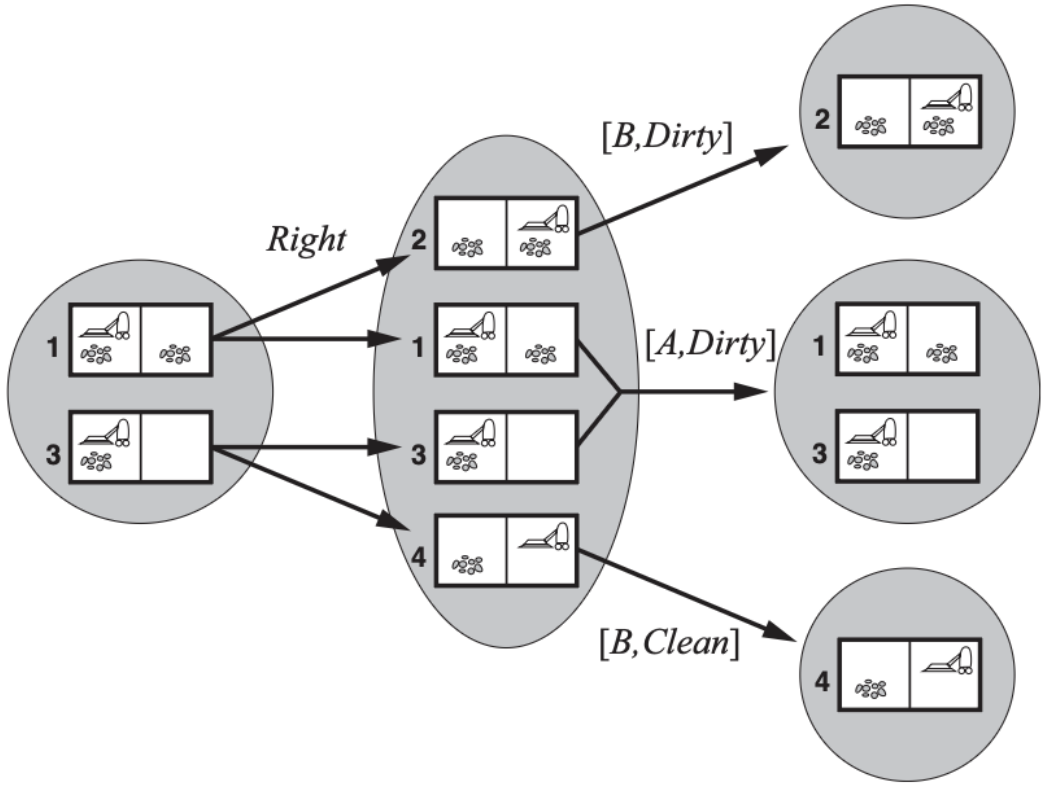
- **Transition model:**

$$\mathit{RESULT}(b,a) = \{b'': b'' = \mathit{UPDATE}(\mathit{PREDICT}(b,a),o) \text{ and } o \text{ in } \mathit{POSIBLE\text{-}PERCEPTS}(\mathit{PREDICT}(b,a))\}$$

Transition in partially observable environments

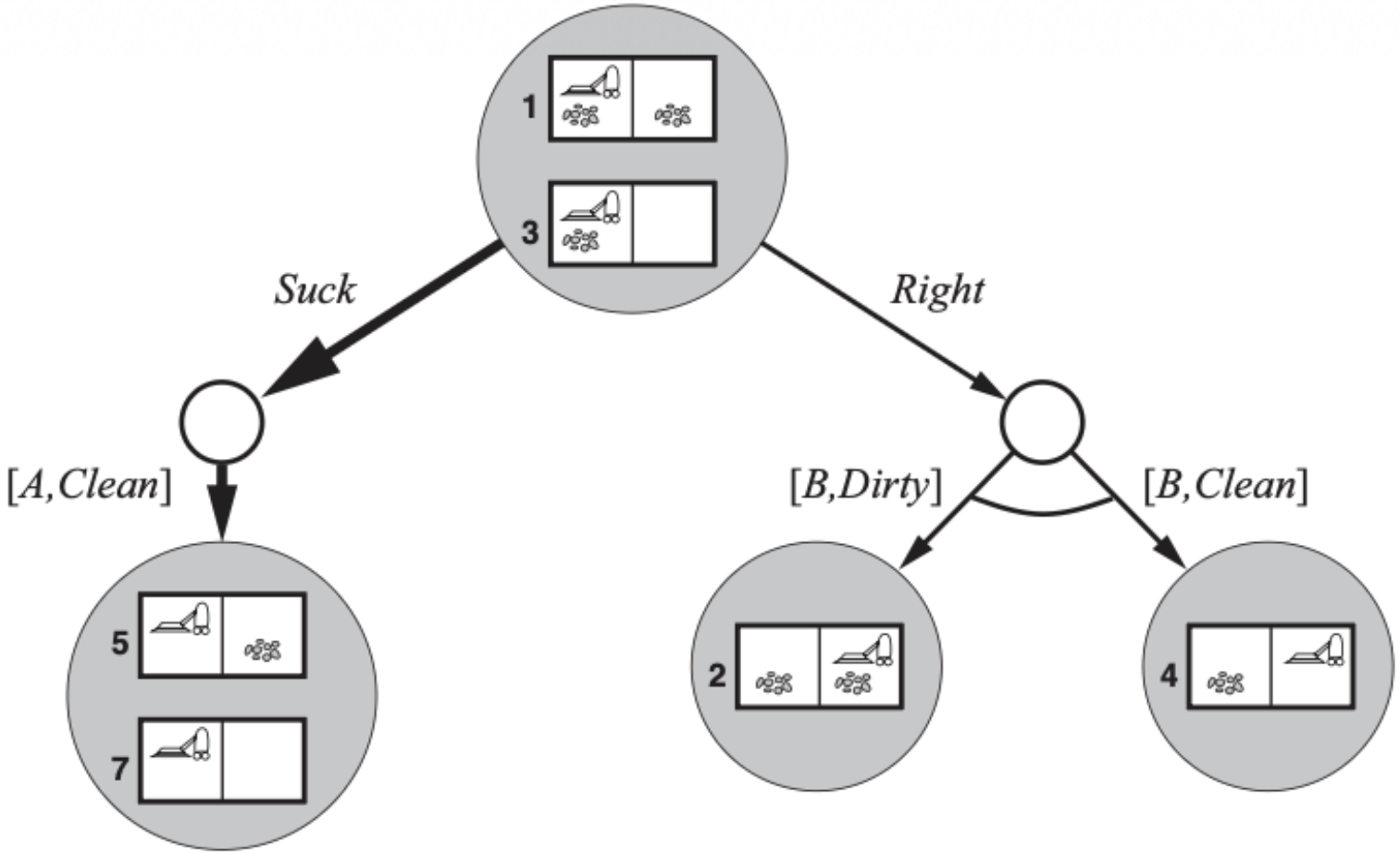


Deterministic



Non-deterministic

Searching in partially observable environments



A solution should be a condition plan rather than a sequence

Updating belief states

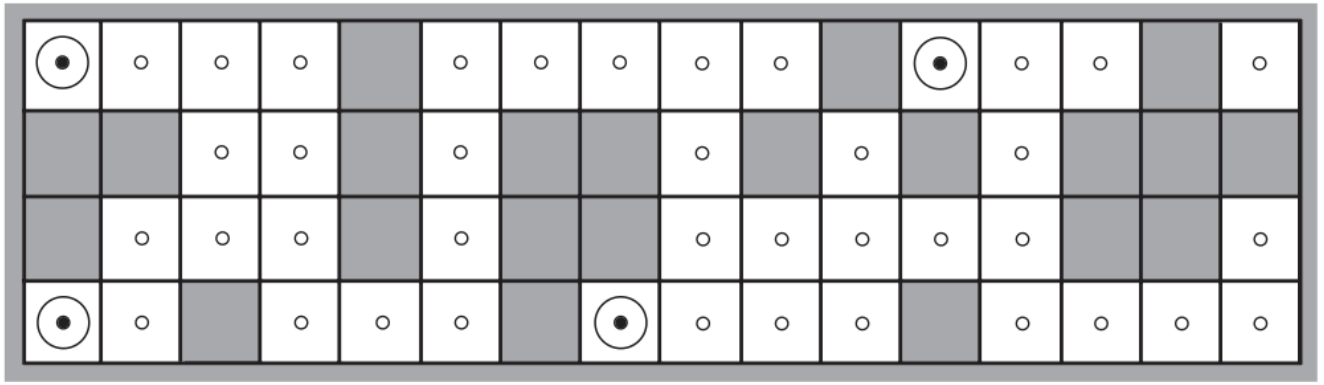
Updating this belief state over time using actions and percepts is essential for making correct decisions.

Updating strategy: b' = *UPDATE*(*PREDICT*(b, a),o)

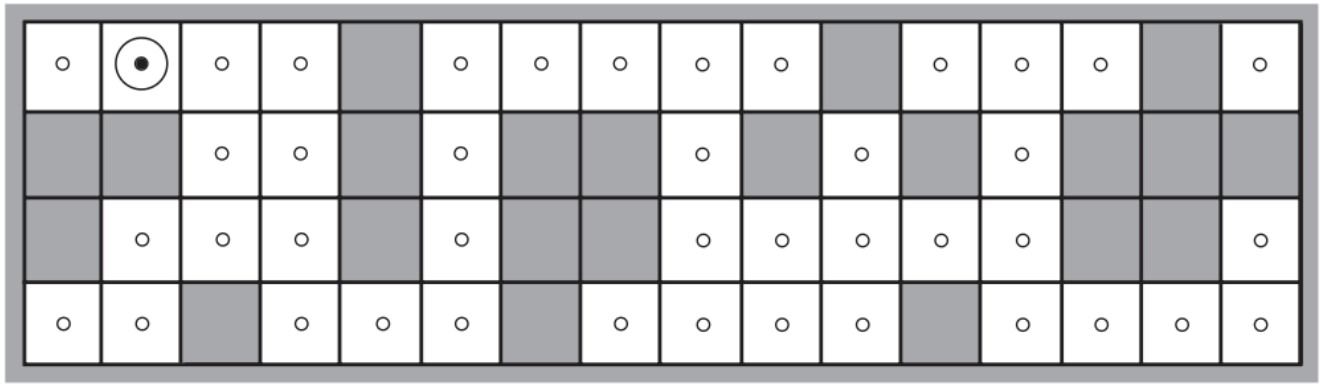
Example of a robot localization.

- A robot doesn't know where it is.
- It has 4 sensors to know whether there is an obstacle.
- Its movements are nondeterministic.

Where is the robot?



(a) Possible locations of robot after E_1 = NSW



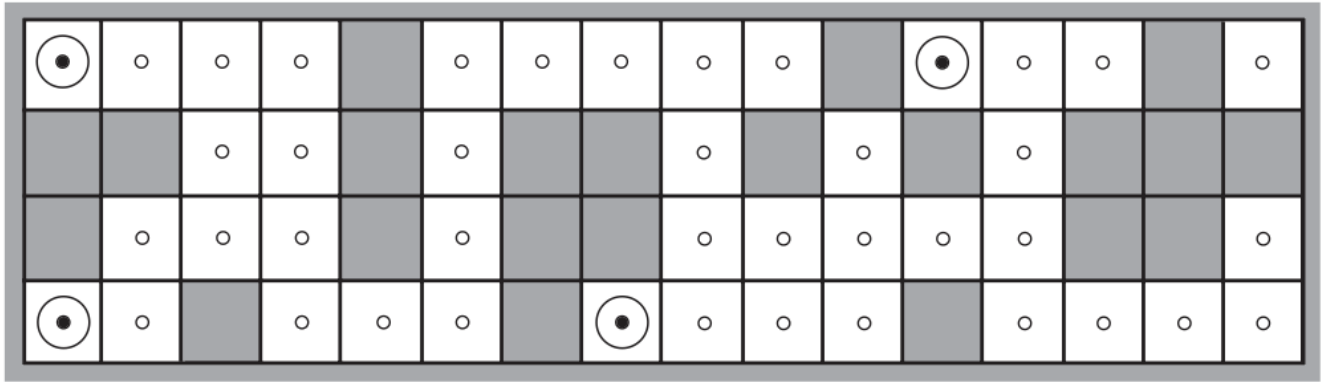
(b) Possible locations of robot After E_1 = NSW, E_2 = NS

Updating belief states

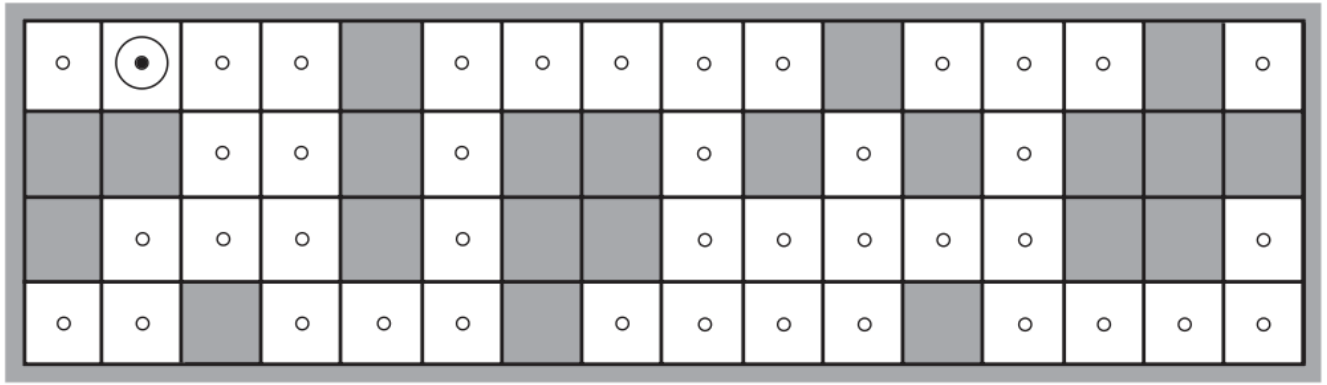
Initially, the robot perceives E1 = ESW, so there are 4 possible locations (belief state).

After moving, the belief state is non-decreasing.

After perceiving E2 = NS, the robot knows its exact location.



(a) Possible locations of robot after E_1 = NSW



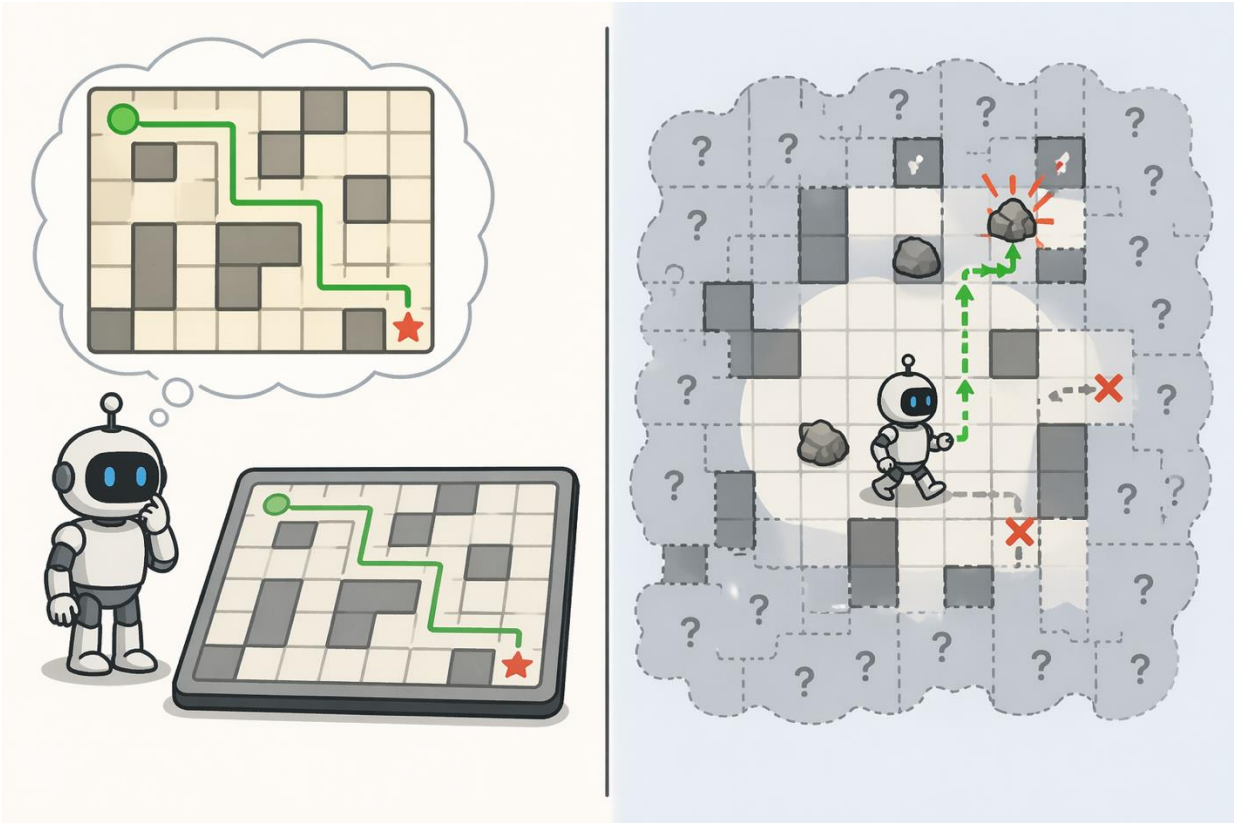
(b) Possible locations of robot After E_1 = NSW, E_2 = NS

4. Online search problems

Offline search: A complete solution is generated before anything changes in the physical world.

Online search:

- Combines computation and action
- Must solved by an agent executing actions
- Useful for dynamic environments
- Useful for nondeterministic environments, since it allows the agent to focus on contingencies that actually arise – not just those that might arise
- Useful for unknown environments

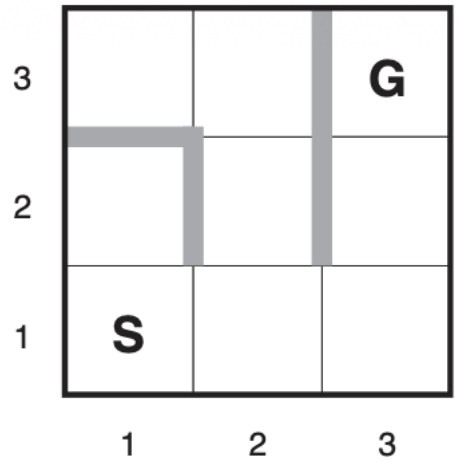


An agent only *know*:

- **Actions(*s*)**: list of actions allowed in a state *s.*
- **Step-cost function c(*s,a,s'*):** can only be determined after *s'* is explored.
- **Goal-test(*s*):** a state is goal or not.

The agent *cannot* determine RESULT(*s,a*) except by actually being in *s* and doing *a.*

The agent *might know* admissible heuristic function h(s).



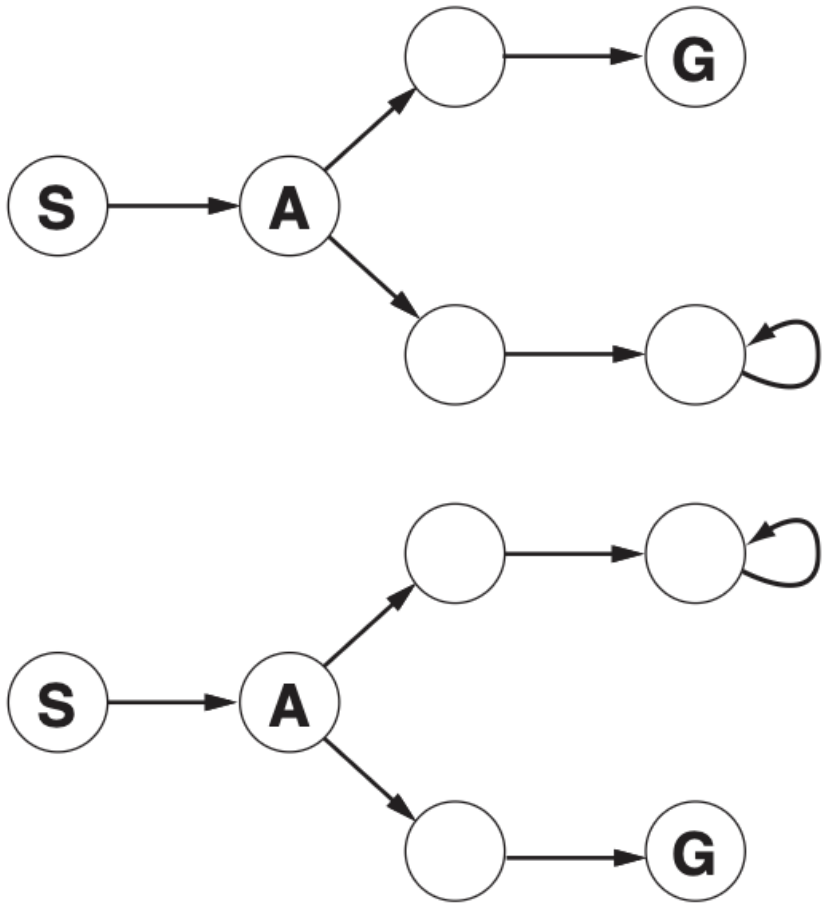
Cost: total path cost the agent actually travels.

Competitive ratio: ratio of actual cost to optimal cost.

The competitive ratio can be infinity because of:

- Irreversible actions: once taken, cannot go back, e.g., fall off a cliff!
- Dead-end state: no path to any goal, e.g., locked in a freezer.

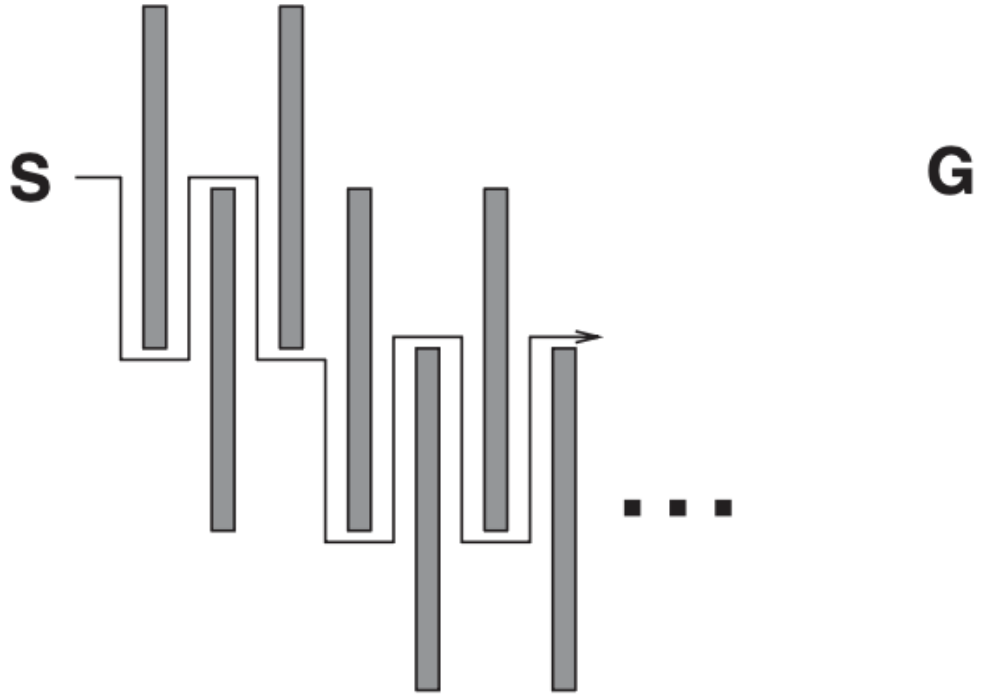
⇒ Limitations: no online algorithm can avoid this in all environments!!!



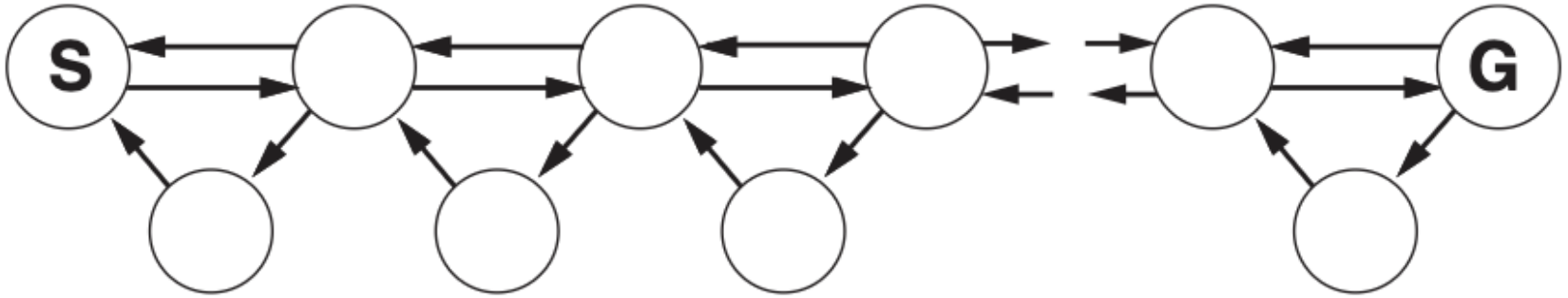
Assumption:

we simply assume that the state space is **safely explorable –** that is, some goal state is reachable from every reachable state.

For example: 8-puzzles.



However, even in safely explorable environments, **no bounded competitive ratio** can be guaranteed if there are paths of unbounded cost.



The agent can only explore from the current state

- Can't bounce around to other search paths, like offline A*
- DFS can be used, as long as actions are *reversible*
- Also, online versions of iterative deepening will work
- Random walk is possible
  - Select at random an available action from current state
  - Preference to previously unexplored actions
  - Will eventually find goal, if space is finite
  - But slow – exponential search in worst case

Exercises

Given a search problem with non-deterministic environments as follow:

Initial state: S

Set of actions:

- R1: S → A, B, C
- R2: A → D
- R3: B → E, F
- R4: C → D, G
- R5: F → H

Goal states: D, G, H

a) Build an AND-OR graph

b) Is this problem solvable? If any, find a condition plan.

Trung tâm Học liệu và Dạy học số

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